

YEAR 10 SCIENCE (EXTENSION)
PHYSICAL SCIENCES
REVISION

1. (a) Distinguish between the following terms.
 - (i) Distance and displacement.
 - (ii) Speed and velocity.
 - (b) A man drives his car 8.00×10^2 m due East for 2.10 mins and then turns North, driving 3.00×10^2 m in 1.35 mins before stopping at a petrol station.
 - (i) Calculate the average speed for the entire motion.
 - (ii) Calculate the displacement made from home to the petrol station. Show your working clearly.
 - (iii) Determine the average velocity for the entire motion.
2. A group of students produced the following ticker tape. One dot is recorded every 0.02 seconds.
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- (a) Describe the type of motion shown by the tape.
 - (b) Analyse the tape and calculate a value for the motion shown. Show your working.
3. A car moves off from a standing start with a uniform acceleration. It reaches 20.0 ms^{-1} after 8.00 s before maintaining its speed for another 15.0 s. As it approaches a set of lights, it brakes uniformly to 8.00 ms^{-1} over 12.0 s. It maintains this velocity as it moves through the lights, which turned green as it approached..
 (Assume the movement has occurred on a straight road.)
- (a) Draw a velocity - time graph for the entire motion.
 - (b) By finding the gradient, calculate:
 - (i) the acceleration at the start of the motion.
 - (ii) the deceleration as the car approached the second set of lights.
 - (c) From the graph, determine the displacement of the car over 40.0 s of the motion.
4. A balloon filled with hydrogen gas has a constant velocity of 40.0 ms^{-1} East. How long will it take to go 4.50×10^2 m ?

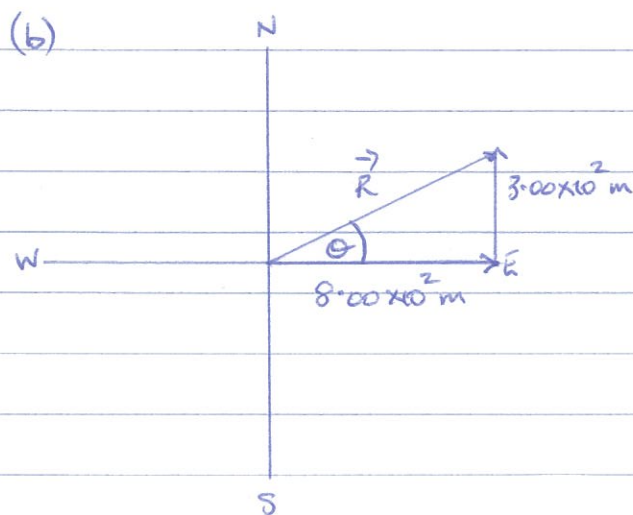
12. During a stunt for a film, a 90.0 kg person leaps from a plane at 5,000 m above the ground. He maintains a streamlined body shape to minimise air resistance. Ignoring any friction, calculate the following.
- (a) How far has the person fallen after 18.0 s?
 - (b) How fast is he travelling at this point?
 - (c) The stunt person deploys a parachute that reduces his velocity to 25.0 ms^{-1} in 90.0 m. Calculate the force exerted by the parachute.
(Hint - calculate acceleration first.)
13. An electron of mass $9.11 \times 10^{-31} \text{ kg}$ is accelerated from rest to $2.45 \times 10^7 \text{ ms}^{-1}$ over a distance of 3.00 cm in an electric field.
- (a) Calculate the magnitude of the acceleration of the electron. Does this surprise you? Comment on this.
 - (b) Consider the size of the force required to accelerate the electron. Are you surprised by its magnitude? Comment on this.

1. (a) (i) **DISTANCE**: how far a body moves in any direction.

DISPLACEMENT: how far a body moves from the start of the motion to the finish of the movement in a particular direction.

(ii) **SPEED**: distance moved per unit time.

VELOCITY: displacement moved in a given direction per unit time.



(i) $\text{speed} = \frac{d}{t}$

$$= \frac{1.1 \times 10^3}{207}$$

$$= \underline{5.31 \text{ ms}^{-1}}$$

(ii) $R = \sqrt{(8.00 \times 10^2)^2 + (3.00 \times 10^2)^2}$

$$= 8.54 \times 10^2 \text{ m}$$

$$\tan \theta = \frac{3.00 \times 10^2}{8.00 \times 10^2}$$

$$\Rightarrow \theta = 20.6^\circ$$

$\therefore \underline{S = 8.54 \times 10^2 \text{ m } \angle 20.6^\circ \text{ N}}$

(iii) $V_{av} = \frac{S}{t}$

$$= \frac{8.54 \times 10^2}{207}$$

$$= \underline{4.13 \text{ ms}^{-1} \angle 20.6^\circ \text{ N}}$$

2. (a) acceleration

(b) $u = \frac{s}{t}$ $v = \frac{s}{t}$

$$= \frac{0.023}{0.08} \quad = \frac{0.082}{0.08}$$

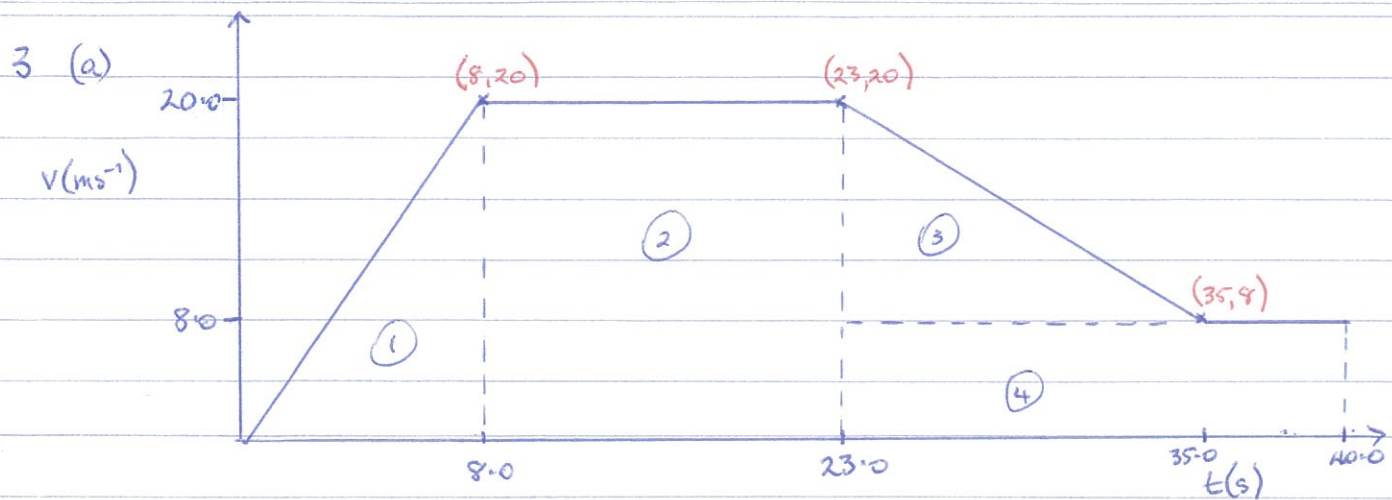
$$= 0.288 \text{ ms}^{-1} \quad = 1.025 \text{ ms}^{-1}$$

$$a = \frac{v-u}{t}$$

$$= \frac{1.025 - 0.288}{0.16}$$

$$= \underline{4.61 \text{ ms}^{-2}}$$

$$t = 8 \times 0.02 = 0.16 \text{ s}$$



(b) (i) $a = \frac{(20-0)}{(8.0-0)} = 2.5 \text{ ms}^{-2}$ forwards

(ii) $a = \frac{(8-20)}{(35-23)} = -1.0 \text{ ms}^{-2}$ backwards.

(c) $s = \text{area under graph}$
 $= \frac{1}{2}(8.0)(20.0) + (15.0)(20.0) + \frac{1}{2}(12.0)(12.0) + (17.0)(8.0)$
 $= 588 \text{ m}$

4. $V = 40.0 \text{ ms}^{-1}$
 $s = 4.50 \times 10^2 \text{ m}$
 $t = ?$

$t = \frac{s}{V}$
 $= \frac{4.50 \times 10^2}{40.0}$
 $= 11.2 \text{ s}$

Forward is +ve.

5. (a) $V = 5.56 \text{ ms}^{-1}$
 $u = 19.4 \text{ ms}^{-1}$
 $a = ?$
 $t = 3.20 \text{ s}$
 $s = ?$

$a = \frac{V-u}{t}$
 $= \frac{5.56-19.4}{3.20}$
 $= -4.32 \text{ ms}^{-2}$ backwards

$\Sigma F = ma$
 $= (1.53 \times 10^3)(-4.32)$
 $= -6.61 \times 10^3 \text{ N}$

$\therefore F = 6.61 \times 10^3 \text{ N backwards}$

(b)

$$v = 5.56 \text{ ms}^{-1}$$

$$u = 19.4 \text{ ms}^{-1}$$

$$a = -4.32 \text{ ms}^{-2}$$

$$t = 3.20 \text{ s}$$

$$s = ?$$

$$s = ut + \frac{1}{2}at^2$$

$$= (19.4)(3.20) + \frac{1}{2}(-4.32)(3.20)^2$$

$$= \underline{10.0 \text{ m forwards}}$$

- (c)
- Hat has inertia - keeps moving forward at 70.0 kmh^{-1} .
 - Flies forward off the back shelf.

6. (a) MASS: a measure of the amount of matter in an object. eg. 10 kg
 WEIGHT: force due to gravity. e.g. 98.0 N near Earth's surface.

(b)

$$F_w = mg$$

$$= (4.50 \times 10^2)(22.5)$$

$$= \underline{1.01 \times 10^4 \text{ N}}$$

- (c) There is always a force due to gravity affecting an object, no matter where it is in the universe.

The astronaut "feels weightless" because he does not experience a reaction force. ie. $R = 0$.



(a)

$$\Sigma F = ma$$

$$\Rightarrow (1.50 \times 10^2 - 80.0) = (70.0)a$$

$$\Rightarrow \underline{a = 1.00 \text{ ms}^{-2} \text{ forwards}}$$



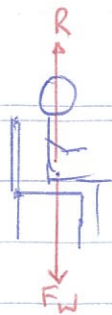
if $F_{\text{forward}} = F_{\text{friction}}$ then

$$\Sigma F = 0$$

$$\Rightarrow a = 0$$

ie. Move at constant velocity.

8.



- Gravity pulls the person down onto the chair.
- The chair exerts an equal and opposite reaction force upwards.
- As they are equal and opposite, the chair supports the person.

9. (a) $2.93 \times 10^{-5} \text{ m}$ (b) $3.652 \times 10^3 \text{ kg}$ (c) $8.316 \times 10^3 \text{ s}$
 (d) $4.95 \times 10^{11} \text{ m}$ (e) $1.944 \times 10^{-1} \text{ ms}^{-1}$ (f) $7.5 \times 10^{-5} \text{ kg}$

10. 1.4×10^2

11. (a) SCALAR: has size (magnitude) only
 VECTOR: has size and direction.
- (b) mass, time, length.
 velocity, acceleration, force

12. (a) $v = ?$

$u = 0 \text{ ms}^{-1}$

$a = 9.80 \text{ ms}^{-2}$

$t = 18.0 \text{ s}$

$s = ?$

↓ +ve

$$s = ut + \frac{1}{2}at^2$$

$$= 0 + \frac{1}{2}(9.80)(18.0)^2$$

$$= \underline{1.59 \times 10^3 \text{ m down.}}$$

(b)

↓ +ve

$$v = u + at$$

$$= 0 + (9.80)(18.0)$$

$$= \underline{1.76 \times 10^2 \text{ ms}^{-1} \text{ down.}}$$

(c) $v = 25.0 \text{ ms}^{-1}$

$u = 1.76 \times 10^2 \text{ ms}^{-1}$

$a = ?$

$t = ?$

$s = 90.0 \text{ m}$

↓ +ve

$$v^2 = u^2 + 2as$$

$$\Rightarrow a = \frac{v^2 - u^2}{2s}$$

$$= \frac{(25.0)^2 - (1.76 \times 10^2)^2}{2(90.0)}$$

$$= -1.69 \times 10^2 \text{ ms}^{-2}$$

$$\Sigma F = ma = (90.0)(-1.69 \times 10^2)$$

$$= -1.52 \times 10^4 \text{ N}$$

$\therefore F = \underline{1.52 \times 10^4 \text{ N upwards}}$

Take forwards as +ve.

13. (a) $v = 2.45 \times 10^7 \text{ ms}^{-1}$

$$u = 0 \text{ ms}^{-1}$$

$$a = ?$$

$$t = ?$$

$$s = 0.0300 \text{ m}$$

$$v^2 = u^2 + 2as$$

$$\Rightarrow a = \frac{v^2 - u^2}{2s}$$

$$= \frac{(2.45 \times 10^7)^2 - 0}{2(0.0300)}$$

$$= \underline{1.00 \times 10^{16} \text{ ms}^{-2} \text{ forwards}}$$

This is a massive acceleration, but the mass is very small so it is possible.

(b)

$$\Sigma F = ma$$

$$= (9.11 \times 10^{-31})(1.00 \times 10^{16})$$

$$= 9.11 \times 10^{-15} \text{ N}$$

This is incredibly small, but when applied to such a small mass, it has a large effect (creates a massive acceleration).